

Raman and SERS characterisation of EV Acacia and EV nano Acacia

Plant extracellular vesicles vs plant-derived nanovesicles: background, experimental conditions and data interpretation

Spontaneous Raman on CaF_2 and SERS on silver-decorated silicon nanowires (AgSiNW),
532 nm

All spectral results are from raw data (no smoothing, no baseline correction)

September 2026

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Summary

Two vesicle preparations called **EV Acacia** and **EV nano Acacia** were received from another laboratory. The isolation protocol, plant species and plant organ were not provided. They were measured by spontaneous Raman spectroscopy on CaF_2 and by surface-enhanced Raman spectroscopy (SERS) on AgSiNW substrates. The main findings are:

- **EV Acacia on CaF_2** gives a reproducible fingerprint across five replicate spectra ($r = 0.96\text{--}0.99$). It is dominated by CH_2/CH_3 deformation at $\sim 1456\text{ cm}^{-1}$, an ester $\text{C}=\text{O}$ stretch at $\sim 1729\text{ cm}^{-1}$ and CH stretching at $\sim 2937\text{ cm}^{-1}$. Weaker protein (amide III, Phe/Tyr) and polysaccharide bands are also present. This is a lipid- and ester-rich vesicle signature.
- **EV nano Acacia on CaF_2** has a different fingerprint. Its strongest band is at 1096 cm^{-1} ($\text{C}-\text{O}/\text{C}-\text{C}$ of carbohydrate, PO_2^- , gauche $\text{C}-\text{C}$), with a saccharide $\text{C}-\text{O}-\text{C}$ band at 900 cm^{-1} and a CH band at $\sim 2891\text{ cm}^{-1}$. The ester $\text{C}=\text{O}$ band is **absent**. Its correlation with the EV Acacia mean is only $r = 0.77$, against $0.96\text{--}0.99$ between EV Acacia replicates.
- **On AgSiNW (SERS)** neither sample reproduces its CaF_2 fingerprint. The 1456 and 1726 cm^{-1} bands were never found together in any of the 100 EV Acacia map spectra. SERS spectra of both samples are dominated by the Si phonon (519 cm^{-1}), Phe (1002 cm^{-1}), carboxylate and broad amorphous-carbon D/G bands (~ 1380 and $\sim 1580\text{--}1607\text{ cm}^{-1}$).
- This is explained by SERS hot-spot and orientation selectivity, a roughly 600-fold smaller photon budget in the AgSiNW settings, and photo-induced carbon on silver. It is **not** evidence that lipids are absent. CaF_2 is therefore the reliable basis for comparing the two samples.
- The CaF_2 difference fits the literature distinction (Section 1). True EVs are secreted, membrane-rich vesicles. Homogenisation-derived nanovesicles carry co-isolated biomass material; for Acacia this is abundant polysaccharide gums and polyphenols. This remains a hypothesis until the provider lab confirms the isolation methods.

1. Background

1.1 Plant extracellular vesicles vs plant-derived nanovesicles

Plants release membrane vesicles, and vesicle-like particles can also be made from plant tissue. Both are often called "plant exosomes", but the literature now separates two different materials [1–3]:

- **Plant-derived extracellular vesicles (PDEVs, "true" plant EVs).** Living cells actively secrete these into the extracellular space (apoplast). They are usually isolated from apoplastic washing fluid (vacuum infiltration followed by low-speed centrifugation), or from the medium of cell-suspension or callus cultures, then purified by ultracentrifugation, density gradients or size-exclusion chromatography [2,4,5]. Their cargo is loaded selectively and includes stress- and defence-related proteins, small RNAs and membrane lipids [1,4]. They take part in plant immunity and cross-kingdom RNA communication with pathogens [1].
- **Plant-derived nanovesicles / nanoparticles (PDNVs, PDNPs, "exosome-like nanoparticles", ELNs).** These form when tissue is juiced, blended or homogenised, and are then collected by differential centrifugation. Membranes of many compartments (plasma membrane, vacuole, organelles) break and reseal. The particles therefore reflect the bulk lipid and metabolite composition of the biomass rather than a selectively loaded cargo [1,3]. Yield is high, which makes them attractive for applications. However, they can be contaminated by co-isolated soluble material and intracellular membranes [2,3].

Pinedo et al. pointed out that the two terms are often used interchangeably. They called for rigorous reporting of the source and isolation method, because these determine what the particles are [2]. Table 1 summarises the differences.

Property	Plant EVs (PDEVs)	Plant nanovesicles (PDNVs / ELNs)
Origin	Actively secreted by living cells into the apoplast	Formed from membranes broken during tissue disruption
Typical source	Apoplastic washing fluid; cell-culture or callus medium	Juice or homogenate of fruit, root, rhizome, leaf
Cargo	Selective: defence proteins, sRNA, membrane lipids	Bulk: reflects the lipid and metabolite composition of the tissue
Yield / purity	Low yield, higher purity	High yield, more co-isolated proteins, metabolites and polysaccharides
Main research use	Plant immunity, cross-kingdom RNA transfer, engineered carriers	Oral delivery, gut / anti-inflammatory therapy, drug carriers, cosmetics
Key references	[1,2,4,5]	[1,3,6–9]

Table 1. Plant EVs and plant-derived nanovesicles compared, based on the cited reviews and primary studies.

1.2 Where they are used

Plant EVs are mainly used in fundamental plant biology. Rutter and Innes isolated EVs from the leaf apoplast of *Arabidopsis* and showed that they are enriched in stress-response proteins [4]. EVs are also being developed as biological carriers. Kocholatá et al. compared tobacco EVs from apoplastic fluid, calli and suspension cultures and loaded callus-derived vesicles with siRNA for delivery [5].

Plant nanovesicles are the most widely applied in biomedicine:

- **Oral and gut applications.** Edible exosome-like nanoparticles from grapefruit, ginger, grape and carrot are taken up by intestinal macrophages and stem cells and induce anti-inflammatory and protective genes. Their lipid composition depends on the source, e.g. phosphatidic-acid-rich ginger vs phosphatidylcholine-rich grapefruit particles [6].

- **Inflammatory bowel disease.** Orally given ginger-derived nanoparticles reduced colitis and colitis-associated cancer in mice [7].
- **Anti-inflammatory activity.** Vesicle-like nanoparticles from honey inhibit the NLRP3 inflammasome and protect against liver injury in mice [8].
- **Drug delivery, skin and cosmetics, microbiome.** Reviews describe PDNVs as natural carriers for drugs and nucleic acids, as skin and cosmetic actives, and as modulators of the gut microbiota [1,9].

1.3 Acacia as a source

Acacia nilotica (now *Vachellia nilotica*) is widely used in traditional medicine. Its main phytochemical classes are tannins, flavonoids and other polyphenols, alkaloids, fatty acids and polysaccharide gums. Its extracts show anti-inflammatory, antioxidant, antimicrobial, antidiarrhoeal and hypoglycaemic activity [10,11]. The exact *Acacia* species and organ used for these samples were not provided. The *Acacia* literature is cited here as general context. Our literature search found no published study of EVs or nanovesicles from *Acacia*. Vesicles from this plant are therefore a new subject, and a label-free chemical fingerprint is a useful first characterisation.

1.4 Why this work was done

Raman spectroscopy gives a label-free chemical fingerprint of EVs: lipids, proteins, nucleic acids and carbohydrates in a single measurement. It has been used to characterise and discriminate EV preparations [12,13]. CaF_2 is a standard low-background substrate for biological Raman spectroscopy [14], and SERS can raise sensitivity by orders of magnitude for very small sample amounts [13,15]. The aims of this analysis were therefore:

- to obtain a reliable Raman fingerprint of EV *Acacia* and EV nano *Acacia* on CaF_2 ;
- to test whether SERS on AgSiNW with a very small amount of sample (1 μL drop; 20 ng for EV nano *Acacia*) reproduces this fingerprint, in particular the lipid bands at 1456 and 1726 cm^{-1} ;
- to find out whether the two preparations differ chemically, as the literature would predict for EVs vs nanovesicles.

2. Experimental conditions

The samples were received from another laboratory; the isolation protocol is unknown. All acquisition parameters below were read from the data file names; the software is LabSpec 6 (Horiba, *.l6s files). Codes whose meaning is not stated in the files are marked as interpretations.

Parameter	CaF ₂ (spontaneous Raman)	AgSiNW (SERS)
Samples / spectra	EV Acacia: s1, s4, s5, s8, s9, s10 (6 spectra). EV nano Acacia: s1 (1 spectrum)	EV Acacia: maps m1–m5 (5 × 20 = 100 spectra) + single S1, S2. EV nano Acacia: maps m1, m6a (40 spectra) + single s3, s4, s4a
Deposition	Drop-cast ("dropCaF2")	1 µL drop, measured at the drop centre; EV nano Acacia 20 ng
Laser	532 nm	532 nm
Grating	1800 gr/mm	1800 gr/mm
Objective	100×	50× long-working-distance ("50XLF")
Integration × accumulations	60 s × 4	1 s × 4
Filter / power code	"100" (interpreted: 100 % laser power)	"10" (interpreted: 10 % laser power)
"BC" code	BC50	BC200 (EV nano Acacia s3, s4: BC100)
Spectral range	~200–3200 cm ⁻¹	200–3200 cm ⁻¹ (S1: 400–2000)

Table 2. Measurement conditions taken from the file names. If "100"/"10" are laser-power percentages, each AgSiNW spectrum collects about $60 \text{ s} \times 100 \% / (1 \text{ s} \times 10 \%) = 600$ times fewer photon-seconds than a CaF₂ spectrum. The 50× objective also has a lower collection efficiency than the 100× objective.

2.1 Data processing

- All analysis was done on **raw counts**, with no smoothing, baseline correction or normalisation of the plotted spectra.
- Noise $\sigma = 1.4826 \cdot \text{MAD}(\text{first difference})/\sqrt{2}$ of each spectrum. Peaks: local maxima with prominence $\geq 5\sigma$ and FWHM $\geq 4.5 \text{ cm}^{-1}$; narrow maxima were accepted only with SNR ≥ 7 and are flagged.
- Two independent re-checks were applied to every peak. (1) The raw maximum within $\pm 6 \text{ cm}^{-1}$ must be the same data point and stand $\geq 3\sigma$ above a line drawn between the flank minima. (2) A repeated search with different settings (4σ , $\pm 50 \text{ cm}^{-1}$ window) must find the peak again within $\pm 2 \text{ cm}^{-1}$.
- On AgSiNW a noise-calibrated search was also used. A band had to exceed the 99th percentile that pure noise reaches in a band-free region ($600\text{--}720 \text{ cm}^{-1}$) of the same spectrum. Spike-like maxima (cosmic rays) were kept only if reproduced in another spectrum.
- Bands in two spectra were counted as the same band if within $\pm 8 \text{ cm}^{-1}$. Assignments follow standard biological Raman tables [16–19]. Si bands follow [20] and carbon D/G bands follow [21–24].

3. Results

3.1 EV Acacia: CaF₂ vs AgSiNW

On CaF₂ all five replicate spectra (s4–s10) passed the 1456 cm⁻¹ band in 10 of 10 detection checks (SNR 18–38, position 1454–1457 cm⁻¹). The ~1728 cm⁻¹ C=O band passed 9–10 checks (SNR 10–20, 1726–1730 cm⁻¹). Spectrum s1 was chosen for the slides (Figure 1, Table 3). Of the two AgSiNW single spectra, S2 reproduced the most CaF₂ bands (2 of 7) and was chosen.

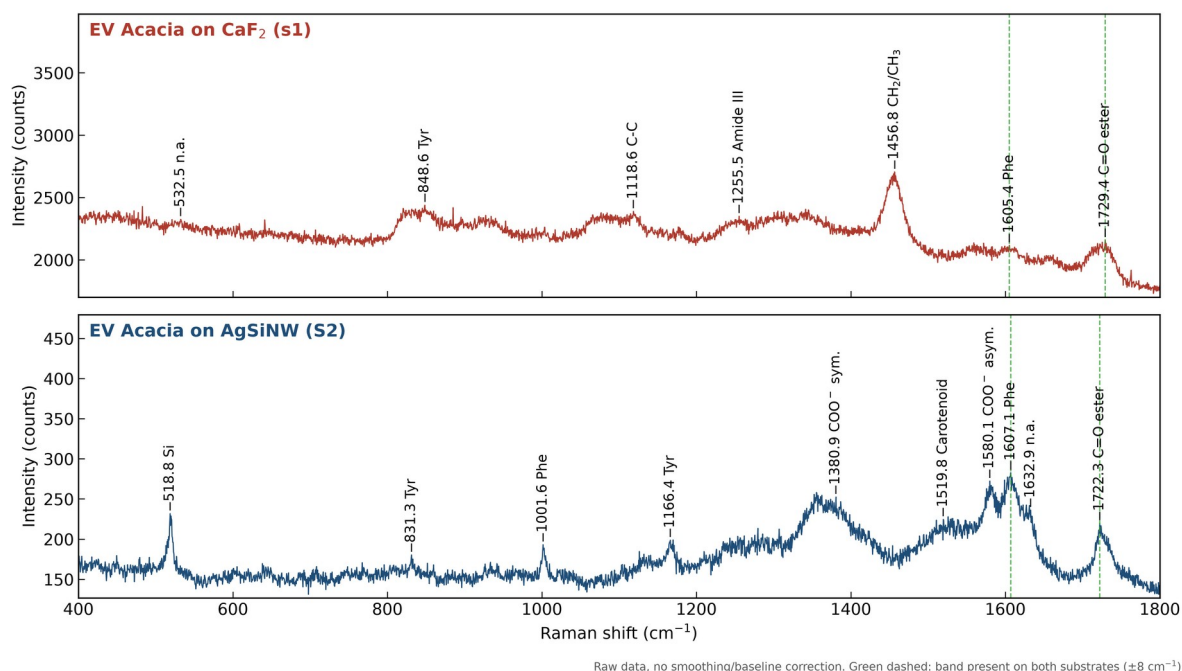


Figure 1. EV Acacia on CaF₂ (s1, top) and on AgSiNW (S2, bottom), raw counts, 400–1800 cm⁻¹. Green dashed lines mark bands present on both substrates (± 8 cm⁻¹).

CaF ₂ (cm ⁻¹)	AgSiNW (cm ⁻¹)	Status	Assignment
–	518.8	only AgSiNW	Si optical phonon (substrate)
532.5	–	only CaF ₂	unassigned
–	831.3	only AgSiNW	Tyr Fermi doublet / O–P–O stretch
848.6	–	only CaF ₂	Tyr ring / C–C (proline), polysaccharide C–O–C
–	1001.6	only AgSiNW	Phe ring breathing
1118.6	–	only CaF ₂	C–C stretch (trans lipid) / C–N (protein)
–	1166.4	only AgSiNW	Tyr C–H bend
1255.5	–	only CaF ₂	Amide III
–	1380.9	only AgSiNW	COO ⁻ sym. stretch / CH ₃ bend / carbon D band
1456.8	–	only CaF ₂	CH ₂ /CH ₃ deformation (lipid, protein)
–	1519.8	only AgSiNW	Carotenoid-type C=C stretch
–	1580.1	only AgSiNW	COO ⁻ asym. stretch / carbon G band / Phe
1605.4	1607.1	in both	Phe/Tyr ring C=C (on AgSiNW overlaps G band)
–	1632.9	only AgSiNW	unassigned
1729.4	1722.3	in both	C=O stretch (ester, lipids)

Table 3. Confirmed bands of EV Acacia, 400–1800 cm⁻¹ (raw data). Green rows: band found on both substrates.

Search of the 100 AgSiNW map spectra for 1456 and 1726 cm⁻¹. The 1456 cm⁻¹ band reached SNR ≥ 5 in 7 spectra and the 1726 cm⁻¹ band in 6 spectra, but **no spectrum showed both at SNR**

≥ 5 . The best pixel (m5, spectrum 12) reached only SNR 5.5 and 4.3. On CaF_2 the same bands have SNR 18–38 and 10–20. The C=O band appears at 1722 cm^{-1} in S2, but without the 1456 cm^{-1} band.

Carbon D/G bands. Fits of the $1200\text{--}1700\text{ cm}^{-1}$ region of the SERS spectra (Table 4) show a broad D band ($1360\text{--}1378\text{ cm}^{-1}$, FWHM $89\text{--}227\text{ cm}^{-1}$), a G band ($1605\text{--}1608\text{ cm}^{-1}$) and a D3 band ($\sim 1545\text{ cm}^{-1}$). This is the typical signature of disordered or amorphous carbon [21–24]. None of these bands appears on CaF_2 .

Spectrum	D (cm^{-1})	D FWHM	G (cm^{-1})	G FWHM	D3 (cm^{-1})	I_D/I_G	R^2
EV Acacia, mean of 100 map spectra	1360.3	227	1604.9	64	1543.8	0.73	0.96
EV Acacia S1	1377.7	89	1608.3	75	1550.0	0.26	0.94
EV Acacia S2	1366.6	96	1607.4	50	1547.6	0.86	0.87
EV nano Acacia s3	1337.3	–	1620.0	–	–	1.52	0.75
EV nano Acacia s4	1385.1	–	1608.2	–	–	0.60	0.91
EV nano Acacia s4a	1384.4	–	1614.4	–	–	0.61	0.94
EV nano Acacia, mean map m1	1381.5	–	1612.0	–	–	0.60	0.95

Table 4. Carbon D/G band fits of the SERS spectra (Lorentzian D, G and Gaussian D3 on a linear background).

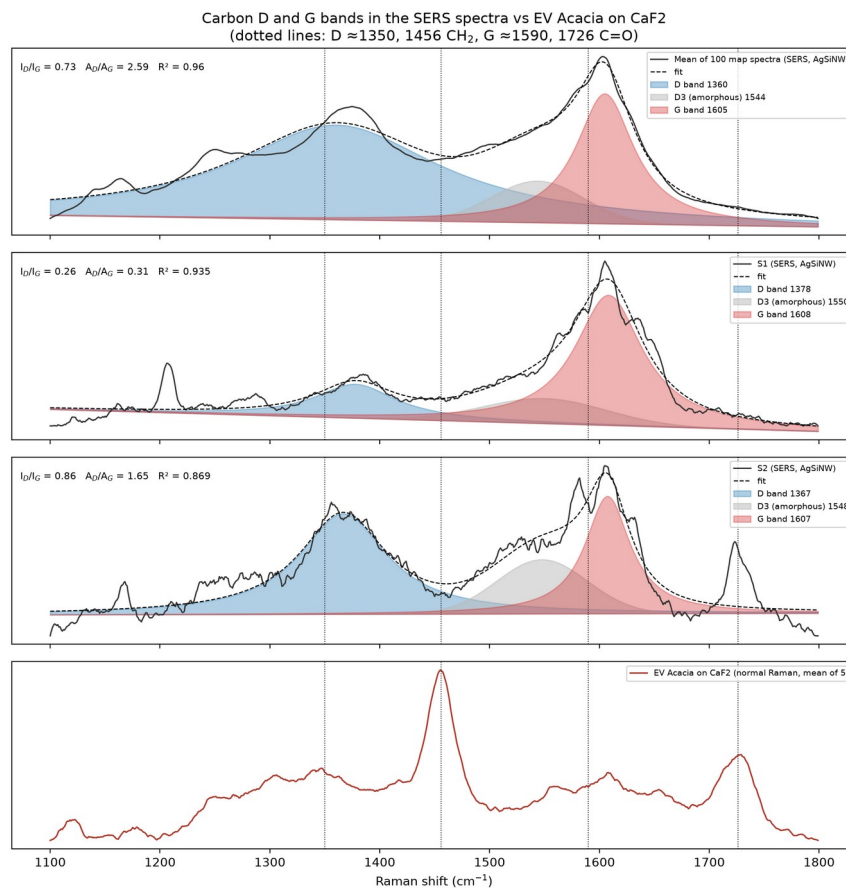
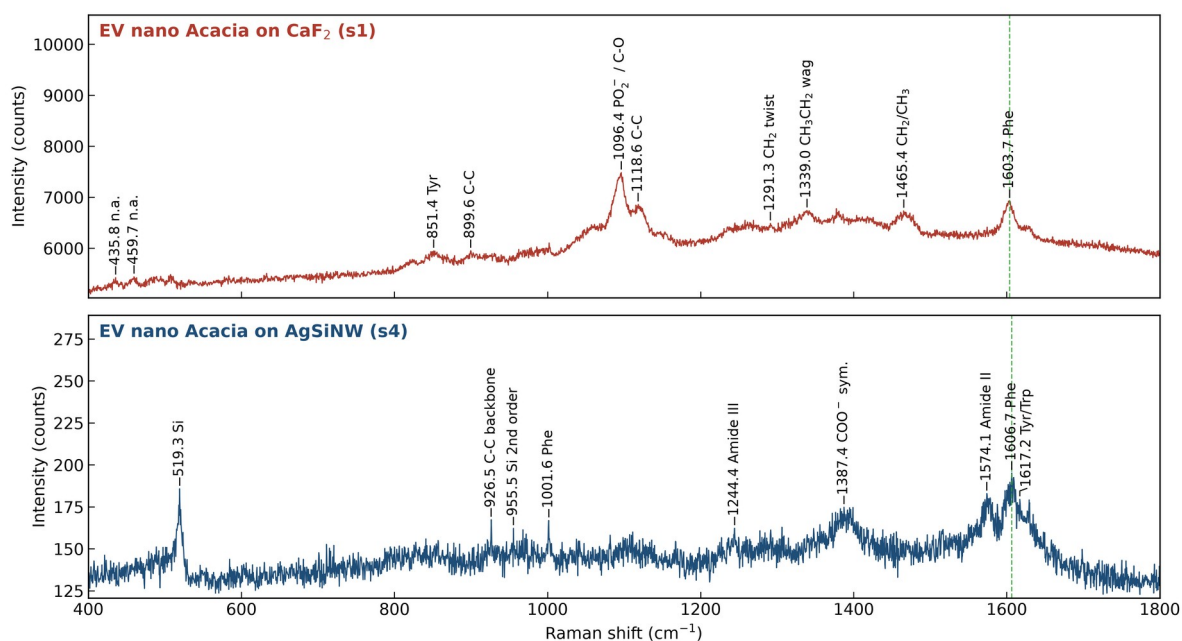


Figure 2. D/G band fits for EV Acacia on AgSiNW compared with the CaF_2 spectra.

3.2 EV nano Acacia: CaF_2 vs AgSiNW

Only one CaF_2 spectrum (s1) of EV nano Acacia was available. Of the three AgSiNW single spectra, s4 matched the most CaF_2 bands, but this was only 1 of 10 (Figure 3, Table 5). The 1456

and 1726 cm^{-1} bands were not detected in any EV nano Acacia spectrum on AgSiNW (maximum SNR 3.4 for 1456 and 3.7 for 1726 cm^{-1} over all 43 spectra).



Raw data, no smoothing/baseline correction. Green dashed: band present on both substrates ($\pm 8 \text{ cm}^{-1}$).

Figure 3. EV nano Acacia on CaF_2 (s1, top) and on AgSiNW (s4, bottom), raw counts, 400–1800 cm^{-1} .

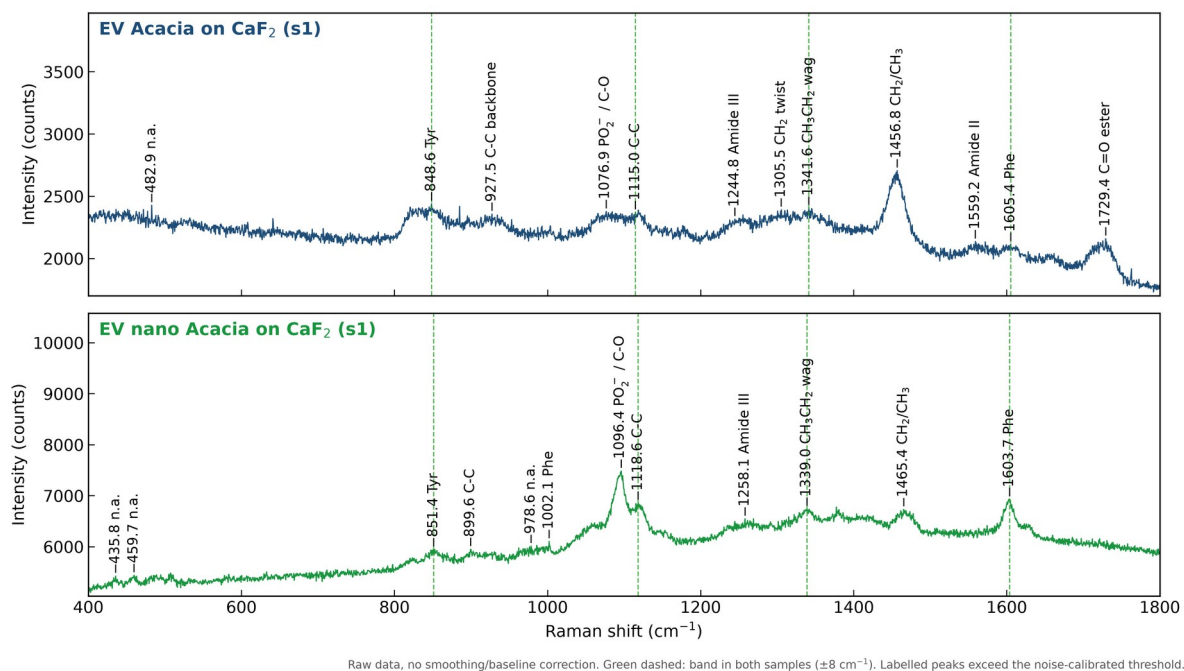
CaF_2 (cm^{-1})	AgSiNW (cm^{-1})	Status	Assignment
435.8	–	only CaF_2	unassigned (skeletal deformation region)
459.7	–	only CaF_2	unassigned (skeletal deformation region)
–	519.3	only AgSiNW	Si optical phonon (substrate)
851.4	–	only CaF_2	Tyr ring / C–C (proline), polysaccharide C–O–C
899.6	–	only CaF_2	C–C / C–O–C (saccharide)
–	926.5	only AgSiNW	C–C stretch, protein backbone
–	955.5	only AgSiNW	Si 2nd-order phonon (substrate)
–	1001.6	only AgSiNW	Phe ring breathing
1096.4	–	only CaF_2	C–O/C–C (carbohydrate), PO_2^- sym., gauche C–C (strongest band, SNR 46)
1118.6	–	only CaF_2	C–C stretch (trans lipid) / C–N (protein)
–	1244.4	only AgSiNW	Amide III
1291.3	–	only CaF_2	CH_2 twist (lipid)
1339.0	–	only CaF_2	CH_3CH_2 wagging (protein, nucleic acid)
–	1387.4	only AgSiNW	COO^- sym. stretch / carbon D band
1465.4	–	only CaF_2	CH_2/CH_3 deformation
–	1574.1	only AgSiNW	Amide II / Trp / carbon G band region
1603.7	1606.7	in both	Phe/Tyr ring C=C (on AgSiNW overlaps G band)
–	1617.2	only AgSiNW	Tyr/Trp ring C=C

Table 5. Confirmed bands of EV nano Acacia, 400–1800 cm^{-1} (raw data).

3.3 EV Acacia vs EV nano Acacia on CaF_2

This is the most reliable comparison, because both samples were measured with identical settings on a low-background substrate. Bands were searched in all spectra of each sample (six EV Acacia,

one EV nano Acacia) and placed at the strongest reproduced maximum. For this reason, some positions differ by a few cm^{-1} from the single-spectrum lists in Tables 3 and 5.



Raw data, no smoothing/baseline correction. Green dashed: band in both samples ($\pm 8 \text{ cm}^{-1}$). Labelled peaks exceed the noise-calibrated threshold.

Figure 4. EV Acacia (top) and EV nano Acacia (bottom) on CaF_2 , raw counts, 400–1800 cm^{-1} .

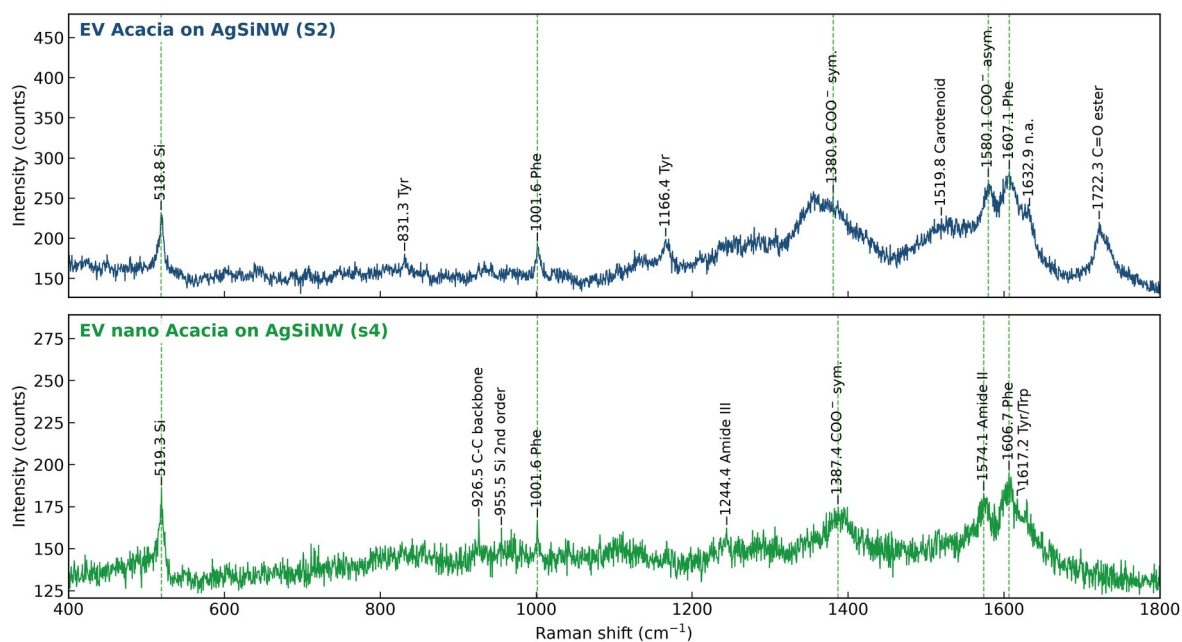
EV Acacia (cm^{-1})	EV nano (cm^{-1})	Status	Assignment
–	435.8	only nano	unassigned
–	459.7	only nano	unassigned
482.9	–	only Acacia	unassigned
848.6	851.4	in both	Tyr ring / C–C (proline), polysaccharide C–O–C
–	899.6	only nano	C–C / C–O–C (saccharide)
927.5	–	only Acacia	C–C stretch, protein backbone
–	978.6	only nano	unassigned
–	1002.1	only nano	Phe ring breathing
1076.9	–	only Acacia	PO_2^- sym. / gauche C–C / C–O
–	1096.4	only nano	C–O/C–C (carbohydrate), PO_2^- sym. (strongest nano band)
1115.0	1118.6	in both	C–C stretch (trans lipid) / C–N (protein)
1244.8	1258.1	only one each ($\Delta 13$)	Amide III
1305.5	–	only Acacia	CH_2 twist (lipid)
1341.6	1339.0	in both	CH_3CH_2 wagging (protein, nucleic acid)
1456.8	1465.4	only one each ($\Delta 8.6$)	CH_2/CH_3 deformation
1559.2	–	only Acacia	Amide II / Trp
1605.4	1603.7	in both	Phe/Tyr ring C=C
1729.4	–	only Acacia	C=O stretch (ester, lipids)

Table 6. EV Acacia vs EV nano Acacia on CaF_2 . Only four bands are shared.

Whole-spectrum correlations with the EV Acacia mean spectrum confirm the difference. The EV Acacia replicates give $r = 0.96$ – 0.99 in the fingerprint region and ≥ 0.999 in the CH region (2750 – 3100 cm^{-1}). EV nano Acacia gives $r = 0.77$ and 0.89 . In the CH region the EV Acacia maximum is at

~2937 cm^{-1} (CH_3/CH_2 asymmetric, protein and lipid). The EV nano Acacia maximum is at ~2891 cm^{-1} (CH_2 symmetric stretch or C–H of carbohydrate).

3.4 EV Acacia vs EV nano Acacia on AgSiNW



Raw data, no smoothing/baseline correction. Green dashed: band in both samples ($\pm 8 \text{ cm}^{-1}$). Labelled peaks exceed the noise-calibrated threshold.

Figure 5. EV Acacia (S2, top) and EV nano Acacia (s4, bottom) on AgSiNW, raw counts, 400–1800 cm^{-1} .

EV Acacia (cm^{-1})	EV nano (cm^{-1})	Status	Assignment
518.8	519.3	in both	Si optical phonon (substrate)
831.3	–	only Acacia	Tyr Fermi doublet / O–P–O
–	926.5	only nano	C–C stretch, protein backbone
–	955.5	only nano	Si 2nd-order phonon (substrate)
1001.6	1001.6	in both	Phe ring breathing
1166.4	–	only Acacia	Tyr C–H bend
–	1244.4	only nano	Amide III
1380.9	1387.4	in both	COO [–] sym. stretch / carbon D band
1519.8	–	only Acacia	Carotenoid-type C=C
1580.1	1574.1	in both	COO [–] asym. / carbon G band / Phe
1607.1	1606.7	in both	Phe/Tyr ring C=C / carbon G band
–	1617.2	only nano	Tyr/Trp ring C=C
1632.9	–	only Acacia	unassigned
1722.3	–	only Acacia	C=O stretch (ester, lipids)

Table 7. EV Acacia vs EV nano Acacia on AgSiNW. The shared bands are the substrate, Phe and the carboxylate/carbon region.

On AgSiNW the two samples look more alike than on CaF_2 . The shared bands (Si, Phe, COO[–]/D/G) are mostly substrate, aromatic amino acids adsorbed on silver, and photo-generated carbon. The only lipid marker, C=O at 1722 cm^{-1} , appears in EV Acacia and not in EV nano Acacia. This agrees in direction with the CaF_2 result.

4. Interpretation

4.1 The two preparations are chemically different

The CaF₂ data show that **EV Acacia** is rich in ester-containing lipids (C=O 1729, CH₂/CH₃ 1456, CH₂ twist 1305, CH 2937 cm⁻¹) and also contains protein (amide III 1245–1255, amide II/Trp 1559, Phe/Tyr 1605, C–C 927 cm⁻¹) [16–18]. This is what a membrane vesicle with phospholipid bilayer and associated proteins is expected to show. The narrow band positions and high replicate correlation also show that the preparation is homogeneous.

EV nano Acacia lacks the ester C=O band. Its strongest bands are at 1096 and 900 cm⁻¹, which are typical of C–O/C–C and glycosidic C–O–C vibrations of carbohydrates [19], and at 2891 cm⁻¹ in the CH region. Plant nanovesicles made by homogenisation carry co-isolated biomass material [1–3]. Acacia tissue and exudates are very rich in polysaccharide gums and polyphenols [10,11]. A carbohydrate-dominated spectrum without a strong ester band is therefore **consistent with** a homogenate-derived nanovesicle preparation. The 1096 cm⁻¹ band can also contain a phosphate (PO₂⁻) contribution. This interpretation is a hypothesis: the isolation methods are unknown and only one CaF₂ spectrum of EV nano Acacia exists.

4.2 Why the CaF₂ bands (1456, 1726 cm⁻¹) are missing on AgSiNW

- **SERS is selective.** Enhancement is strongest for molecules within a few nanometres of a silver hot spot. It also depends on their orientation relative to the surface (surface selection rules) [15,25,26]. Aromatic side chains (Phe, Tyr) and carboxylates adsorb on silver and dominate. The ester groups and alkyl chains inside a bilayer are further away and are not enhanced.
- **Much smaller signal budget.** 1 s at a 10 % power setting with a 50× LF objective, against 60 s at 100 % with 100×, gives about 600 times fewer photon-seconds before collection-efficiency losses. Bands at SNR 20–38 on CaF₂ fall to the noise level unless they are strongly enhanced.
- **Photo-generated carbon.** Laser heating at silver hot spots degrades organic material to amorphous carbon. This produces broad D (~1360–1385) and G (~1580–1610 cm⁻¹) bands [21–24], which were found in all SERS spectra (Table 4). They overlap the 1380–1610 cm⁻¹ region and raise the background near 1456 cm⁻¹.
- **Substrate and dilution.** Si phonons at 519 and ~955 cm⁻¹ [20] and a very small analyte amount (1 µL; 20 ng of EV nano Acacia) reduce the fraction of the spectrum carrying vesicle information.

The missing bands are therefore a property of the SERS measurement, not proof that the vesicles lack lipids. For compositional comparison the CaF₂ spectra should be used. SERS is useful as a sensitivity test and for protein/aromatic markers.

4.3 What this means for possible uses

If EV Acacia is an apoplastic or secreted EV preparation, its lipid- and protein-rich membrane signature suits studies of plant vesicle biology and use as natural carriers, where membrane integrity matters [1,4,5]. If EV nano Acacia is a homogenate-derived nanovesicle preparation, its carbohydrate and polyphenol load fits the application area where plant nanovesicles are most studied. That area is oral, gut and anti-inflammatory use [6–8], where Acacia's own anti-inflammatory and antioxidant compounds [10,11] could add to the effect. Both statements are conditional on the isolation methods being confirmed.

5. Limitations

- The isolation method, plant species and organ, particle concentration (except 20 ng for EV nano Acacia on AgSiNW) and storage history are unknown.
- EV nano Acacia on CaF₂ is a single spectrum, so its reproducibility cannot be tested.
- No blank-substrate spectra (bare CaF₂, bare AgSiNW) or buffer controls were provided.
- AgSiNW settings differ between spectra (BC100 vs BC200), and power/filter codes were interpreted from file names.
- No particle size, morphology or marker data (NTA, TEM, protein markers) were available, as MISEV2023 recommends [27].
- Band assignments are tentative; several bands (e.g. 1096, 1380, 1605 cm⁻¹) have overlapping contributions.

6. Recommendations

- Ask the provider lab for the isolation protocol (apoplastic wash or homogenisation; ultracentrifugation, SEC or density gradient), species, organ and particle concentration. Report them following MISEV2023 [27] and Pinedo et al. [2].
- Add NTA or DLS (size, concentration), TEM (morphology) and protein quantification for both samples.
- Record more CaF₂ spectra of EV nano Acacia (≥ 5 positions) and bare-substrate and buffer blanks.
- Measure reference compounds on CaF₂ under the same conditions: phosphatidylcholine, gum arabic (Acacia senegal gum) and tannic acid. These test the lipid vs polysaccharide/polyphenol assignment directly.
- For SERS, use lower laser power with longer integration to avoid carbonisation (check for the D/G bands), keep one BC setting, and consider aggregating the vesicles on the substrate or using a spacer-free Ag colloid.

References

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